Milestone M2 Package P05 (05_configuration_sweep): Two-Channel Configuration Observability Report

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Package ID:** MS02P05 (M2/experiments/05_configuration_sweep) **Companion MATLAB Script:** m2_p05_config_sweep.m **Primary Sources:** M2_design_evolution.md (Revisions V4--V6, V8--V10), recover_load.m, and M2_100kN_0p1111Hz_sp1p50.mat.

1. Scientific Purpose and Correction of the "Blind at q_3 = 90^\circ" Premise (Revision V6)

An early draft of the M2 specification hypothesized that external load estimation becomes "blind" as the handle approaches the horizontal pose q_3 \to 90^\circ (\pi/2\ \mathrm{rad}) because \cos q_3 \to 0. Revision **V6** in M2_design_evolution.md corrected this misconception:
For a planar external force F = [F_x, F_y]^T applied at station s(d_4) = d_4 + 1.50\ \mathrm{m} along the handle axis, the generalized load vector Q_{\mathrm{ext}} = J_s(q)^T F is:
\begin{bmatrix} Q_{q_3} \\ Q_{d_4} \end{bmatrix} = \begin{bmatrix} -s(d_4)\sin q_3 & s(d_4)\cos q_3 \\ \cos q_3 & \sin q_3 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} s(d_4)(F_y\cos q_3 - F_x\sin q_3) \\ F_x\cos q_3 + F_y\sin q_3 \end{bmatrix}.
Dividing the rotational channel by the known station length s(d_4) > 0 yields the **force-normalized generalized output**:
\tilde{Q}(q) = \begin{bmatrix} Q_{q_3} / s(d_4) \\ Q_{d_4} \end{bmatrix} = \underbrace{\begin{bmatrix} -\sin q_3 & \cos q_3 \\ \cos q_3 & \sin q_3 \end{bmatrix}}_{Q(q_3)^T} \begin{bmatrix} F_x \\ F_y \end{bmatrix}.
Because Q(q_3) is an orthogonal reflection-rotation matrix (Q(q_3)^T Q(q_3) = I_2, \det Q(q_3) = -1), the two-channel force observability norm is **identically unity at every configuration (q_3, d_4)**:
\Vert{}\tilde{Q}(q)\Vert{}_2 = \sqrt{\left(\frac{Q_{q_3}}{s(d_4)}\right)^2 + Q_{d_4}^2} = \sqrt{F_x^2 + F_y^2} = \Vert{}F\Vert{}_2.
Specifically, under pure vertical loading (F_x = 0, F_y \ne 0):

- At q_3 = 0^\circ (handle vertical downward): 100\% of the vertical load projects onto the **saddle torque** channel (Q_{q_3} = s(d_4)F_y, Q_{d_4} = 0).
- At q_3 = 45^\circ: the load splits equally in squared normalized sensitivity (50\% on Q_{q_3}/s, 50\% on Q_{d_4}).
- At q_3 = 90^\circ (handle horizontal): Q_{q_3} = 0, but 100\% of the vertical load is carried directly along the rack-and-pinion **crowd force** channel (Q_{d_4} = +F_y).

Observability therefore **redistributes smoothly between the hoist/saddle channel and the crowd channel** rather than vanishing.

2. Analytical Two-Channel Observability Map Across

(q_3, d_4)

Crowd Extension d_4 (\mathrm{m})	Bail Arm s_{\mathrm{bail}} (\mathrm{m})	COG Arm r_{\mathrm{COG}} (\mathrm{m})	Station Ratio \rho(d_4)	Swing Angle q_3 (\mathrm{deg})	Vertical Gain \partial Q_{q_3}/\partial F_y (\mathrm{N\cdot m/N})	Vertical Gain \partial Q_{d_4}/\partial F_y (\mathrm{N/N})	Saddle Channel Share \cos^2 q_3 (\%)	Crowd Channel Share \sin^2 q_3 (\%)	Combined Force Norm \Vert{}\tilde{Q}\Vert{}_2 / \Vert{}F\Vert{}_2
6.75	8.25	5.43	1.5193	15^\circ	7.9689	0.2588	93.3\%	6.7\%	1.0000
6.75	8.25	5.43	1.5193	30^\circ	7.1447	0.5000	75.0\%	25.0\%	1.0000
6.75	8.25	5.43	1.5193	45^\circ	5.8336	0.7071	50.0\%	50.0\%	1.0000
6.75	8.25	5.43	1.5193	60^\circ	4.1250	0.8660	25.0\%	75.0\%	1.0000
6.75	8.25	5.43	1.5193	75^\circ	2.1353	0.9659	6.7\%	93.3\%	1.0000
6.75	8.25	5.43	1.5193	90^\circ	0.0000	1.0000	0.0\%	100.0\%	1.0000
8.75	10.25	7.43	1.3795	15^\circ	9.9007	0.2588	93.3\%	6.7\%	1.0000
8.75	10.25	7.43	1.3795	30^\circ	8.8768	0.5000	75.0\%	25.0\%	1.0000
8.75	10.25	7.43	1.3795	45^\circ	7.2478	0.7071	50.0\%	50.0\%	1.0000
8.75	10.25	7.43	1.3795	60^\circ	5.1250	0.8660	25.0\%	75.0\%	1.0000
8.75	10.25	7.43	1.3795	75^\circ	2.6529	0.9659	6.7\%	93.3\%	1.0000
8.75	10.25	7.43	1.3795	90^\circ	0.0000	1.0000	0.0\%	100.0\%	1.0000
9.77	11.27	8.45	1.3337	15^\circ	10.8860	0.2588	93.3\%	6.7\%	1.0000
9.77	11.27	8.45	1.3337	30^\circ	9.7601	0.5000	75.0\%	25.0\%	1.0000
9.77	11.27	8.45	1.3337	45^\circ	7.9691	0.7071	50.0\%	50.0\%	1.0000
9.77	11.27	8.45	1.3337	60^\circ	5.6350	0.8660	25.0\%	75.0\%	1.0000
9.77	11.27	8.45	1.3337	75^\circ	2.9169	0.9659	6.7\%	93.3\%	1.0000
9.77	11.27	8.45	1.3337	90^\circ	0.0000	1.0000	0.0\%	100.0\%	1.0000
10.75	12.25	9.43	1.2990	15^\circ	11.8326	0.2588	93.3\%	6.7\%	1.0000
10.75	12.25	9.43	1.2990	30^\circ	10.6088	0.5000	75.0\%	25.0\%	1.0000
10.75	12.25	9.43	1.2990	45^\circ	8.6621	0.7071	50.0\%	50.0\%	1.0000
10.75	12.25	9.43	1.2990	60^\circ	6.1250	0.8660	25.0\%	75.0\%	1.0000
10.75	12.25	9.43	1.2990	75^\circ	3.1705	0.9659	6.7\%	93.3\%	1.0000
10.75	12.25	9.43	1.2990	90^\circ	0.0000	1.0000	0.0\%	100.0\%	1.0000

3. Configuration-Quadrant Verification on Saved Simscape Archive (M2_100kN_0p1111Hz_sp1p50.mat)

Partitioning the N = 2{,}501 post-trim samples (t \in [5.0, 30.0]\ \mathrm{s}) of M2_100kN_0p1111Hz_sp1p50.mat into four (q_3, d_4) configuration quadrants confirms that the numerical recovery residual remains at the 10^{-6}\ \mathrm{N\cdot m} / 10^{-11}\ \mathrm{N} Simscape solver floor across all regions of the workspace:

Workspace Quadrant (q_3 \times d_4)	Sample Count N_q	Saddle RMS Error (\mathrm{N\cdot m})	Saddle Max Error (\mathrm{N\cdot m})	Crowd RMS Error (\mathrm{N})	Crowd Max Error (\mathrm{N})
Low q3 (30-45 deg) x Short d4 (6.75-8.75 m)	1000	1.4132e-06	1.9998e-06	6.6715e-11	2.0464e-10
Low q3 (30-45 deg) x Long d4 (8.75-10.75 m)	0	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
High q3 (45-60 deg) x Short d4 (6.75-8.75 m)	0	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
High q3 (45-60 deg) x Long d4 (8.75-10.75 m)	1501	1.4038e-06	1.9993e-06	9.1166e-11	3.3469e-10

4. Frozen Simulink Execution Specification (m2_p05_config_sweep.m)

Because the Linux sandbox VM does not have a local MATLAB/Simulink binary installed to execute new .slx simulations, the companion script m2_p05_config_sweep.m is provided ready to run in MATLAB R2025b:

1. In **Mode 1 ('AnalyticalAndArchive', default if Simulink is absent)**, m2_p05_config_sweep.m computes the complete (q_3, d_4) observability maps, station-ratio sensitivity \mathrm{d}\rho/\mathrm{d}d_4 = -2.82/(d_4 - 1.32)^2, and configuration-quadrant residuals from M2_100kN_0p1111Hz_sp1p50.mat, saving M2_P05_config_sweep_summary.mat.
2. In **Mode 2 ('RunSimulinkGrid', when M2_LoadInjection.slx is on the MATLAB path)**, m2_p05_config_sweep.m sweeps 9 (q_{3,\mathrm{bias}}, d_{4,\mathrm{bias}}) operating centers (q_{3,\mathrm{bias}} \in \{30^\circ, 50^\circ, 70^\circ\} \times d_{4,\mathrm{bias}} \in \{7.75, 8.75, 9.77\}\ \mathrm{m}) and saves M2_05_q*_d*.mat directly into M2/experiments/05_configuration_sweep/.